

A Wearable Sensor Case Study on Human Activity Recognition

Google Data Analytics Capstone Project

Dataset: PAMAP2 Physical Activity Monitoring (UCI Machine Learning Repository)

Prepared by Kaviratna Narayansamy

Contents

Executive Summary	3
1. Ask — Defining the Problem	4
2. Prepare — Getting to Know the Data	4
3. Process — Cleaning the Data	5
4. Analyze — Finding the Story in the Signal	5
4.1 How Intense Is Each Activity?	5
4.2 Which Sensor Location Tells Activities Apart Best?	7
4.3 Consistency vs. Chaos: A Variability Analysis	8
4.4 The Heart Rate Angle	9
4.5 Can a Machine Learn to Tell These Activities Apart?	10
5. Share — Turning Numbers into a Story	12
6. Act — Engineering Recommendations	12
7. Limitations	13
8. Tools & Skills Demonstrated.....	13

Executive Summary

Every time someone sits down, climbs a flight of stairs, or breaks into a jog, their body produces a distinct pattern of motion and physiological response. This project asks a simple question with real engineering consequences: can wearable sensors reliably tell these everyday activities apart, and if so, which sensor placement does the best job?

Using the PAMAP2 dataset — 100Hz motion data from wrist, chest, and ankle-mounted sensors plus heart rate, collected from 9 volunteers — this analysis compares six activities: sitting, standing, walking, running, ascending stairs, and descending stairs. The findings are clear and, in places, genuinely surprising:

- The ankle sensor is the strongest single predictor of activity, its acceleration signal ranges nearly 5x wider across activities than the chest sensor does.
- The chest sensor is almost useless on its own, it barely moves whether a person is sitting or running, because the torso stays relatively stable regardless of activity.
- Motion sensors cannot distinguish sitting from standing but heart rate can, cleanly. The two signals cover each other's blind spots.
- A machine learning model trained on engineered features correctly classified activity 91% of the time, even when tested on people it had never seen during training.

Together, these results suggest a practical engineering recommendation: a single ankle-mounted accelerometer combined with a heart rate monitor covers most of what's needed to distinguish these six activities, with the biggest remaining challenge being telling ascending stairs apart from descending stairs.

1. Ask — Defining the Problem

Wearable devices increasingly promise to track what we do all day, not just how many steps we take. But that promise depends entirely on sensor placement and signal quality. A smartwatch on the wrist "sees" the world very differently than a chest strap or an ankle band — and if a company chooses the wrong placement, their activity-tracking feature simply won't work as advertised.

This project approaches that engineering problem from the data side, using a research-grade multi-sensor dataset to answer questions a wearable design team would actually need answered:

- How does movement intensity differ across six common human activities?
- Which wearable sensor location — wrist, chest, or ankle — provides the strongest distinction between activities?
- How do physiological responses (heart rate) change with movement intensity, and where do they add information that motion sensors miss?
- Which specific signal features are most useful for telling activities apart?
- Do different people show consistent movement patterns for the same activity, or does it vary meaningfully person to person?

2. Prepare — Getting to Know the Data

The PAMAP2 dataset was collected by DFKI (the German Research Center for Artificial Intelligence) using three wireless IMUs (hand, chest, ankle) sampled at 100Hz, plus a chest-worn heart rate monitor sampled at roughly 9Hz. Nine subjects — mostly employees or students — each followed a structured activity protocol.

Table 1. Subject Demographics

Subject	Sex	Age	Height (cm)	Weight (kg)	Resting HR	Max HR	Dominant Hand
101	Male	27	182	83	75	193	Right
102	Female	25	169	78	74	195	Right
103	Male	31	187	92	68	189	Right
104	Male	24	194	95	58	196	Right
105	Male	26	180	73	70	194	Right
106	Male	26	183	69	60	194	Right
107	Male	23	173	86	60	197	Right
108	Male	32	179	87	66	188	Left
109*	Male	31	168	65	54	189	Right

* Subject 109 completed none of the six target activities and is excluded from all analysis below.

The sample skews heavily male (8 of 9 subjects) and young adult (23–32 years) — an important limitation to keep in mind when generalizing any findings beyond this group (see Limitations).

Why Subject 109 Drops Out

This comparison only covers six activities: sitting, standing, walking, running, ascending stairs, and descending stairs. Subject 109 never performed any of those six — their session only covered optional activities like computer work, laundry folding, and soccer. This isn't a data error; that subject simply has no usable rows once the data is filtered to this study's scope, which is why the analysis proceeds with 8 subjects rather than 9.

Is This Data Trustworthy?

The dataset is peer-reviewed, published by an academic research lab, freely available with full documentation, and has become a standard benchmark in activity-recognition research. Its main limitations for this project are age (collected in 2012, so hardware has since evolved) and scope (a controlled lab protocol rather than free-living, everyday conditions).

3. Process — Cleaning the Data

Raw data at 100Hz across 9 subjects totals millions of rows, most of it irrelevant to this study. Four cleaning decisions shaped the final dataset:

Table 2. Activities Included in This Analysis

Activity ID	Activity Name
2	Sitting
3	Standing
4	Walking
5	Running
12	Ascending Stairs
13	Descending Stairs

- Discarded "transient" rows (activityID = 0): the dataset's own documentation labels the gaps between activities — walking to a new location, waiting for instructions — as activity 0, and explicitly recommends discarding them before any analysis.
- Dropped the four orientation columns per sensor: the dataset documentation flags these as invalid for this data collection, so keeping them risks teaching a model to trust broken signals.
- Preferred the $\pm 16g$ accelerometer over the $\pm 6g$ accelerometer: the $\pm 6g$ sensor saturates (maxes out) during high-impact movement like running or landing on a stair, silently corrupting exactly the activities this study cares about most.
- Computed acceleration magnitude — $\sqrt{(x^2 + y^2 + z^2)}$ — for each sensor, turning three direction-dependent values into one number representing total movement intensity, independent of how the sensor happened to be oriented on the body.

4. Analyze — Finding the Story in the Signal

4.1 How Intense Is Each Activity?

Averaging acceleration magnitude across each activity gives a first clean read on movement intensity, and the ankle sensor tells the clearest story:

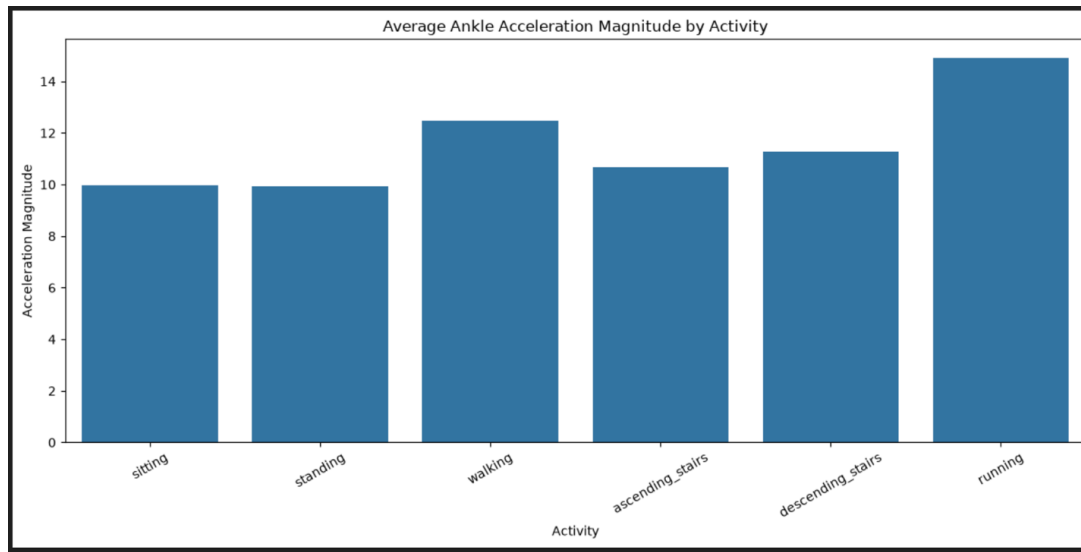


Figure 1. Average ankle acceleration magnitude by activity. A steady climb from static postures through locomotion to running.

Table 3. Average Acceleration Magnitude and Heart Rate by Activity

Activity	Ankle Avg	Chest Avg	Hand Avg	Heart Rate Avg (bpm)
Sitting	9.95	9.80	9.68	80.04
Standing	9.94	9.83	9.70	88.55
Walking	12.46	9.88	10.94	112.79
Ascending Stairs	10.68	9.92	10.29	129.58
Descending Stairs	11.29	9.90	9.67	129.19
Running	14.89	9.95	13.36	156.52

Ankle acceleration scales cleanly with real-world intensity: sitting and standing sit almost on top of each other near 9.9 (essentially just gravity, since the ankle barely moves), walking and the two stair activities cluster in the low-to-mid teens, and running stands clearly apart at the top. Chest, by contrast, barely shifts at all across the entire activity range — a first hint that it's the weakest of the three sensors for this task.

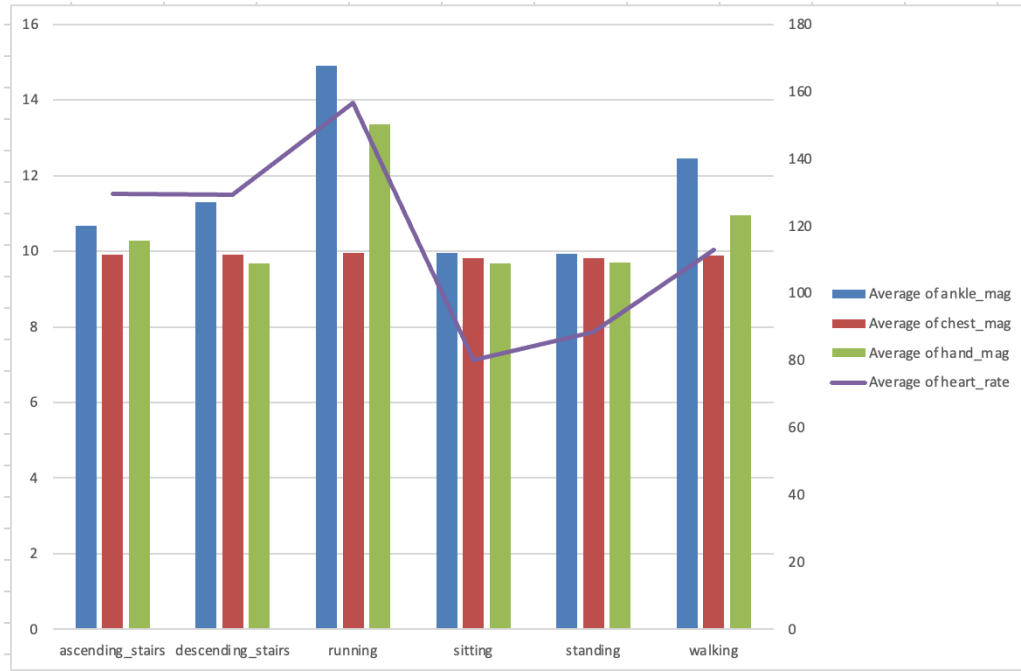


Figure 2. Average sensor magnitude (bars, left axis) and heart rate (line, right axis) by activity. Splitting the axes prevents heart rate's larger scale from hiding the accelerometer differences.

4.2 Which Sensor Location Tells Activities Apart Best?

Averages alone are easy to eyeball but easy to argue with. To make the sensor comparison rigorous, each sensor's range — its highest activity average minus its lowest — quantifies how much "spread" that sensor offers for distinguishing activities:

Table 4. Sensor Discriminative Range

Sensor	Highest Avg (Activity)	Lowest Avg (Activity)	Range
Ankle	14.89 (Running)	9.94 (Standing)	4.95
Hand	13.36 (Running)	9.67 (Descending Stairs)	3.69
Chest	9.95 (Running)	9.80 (Sitting)	0.15

The ankle sensor's range is roughly 33x wider than the chest sensor's — strong, quantified evidence that ankle placement is the superior choice if a wearable can only carry one motion sensor. The boxplots below make the same point visually, showing not just averages but the full spread of values:

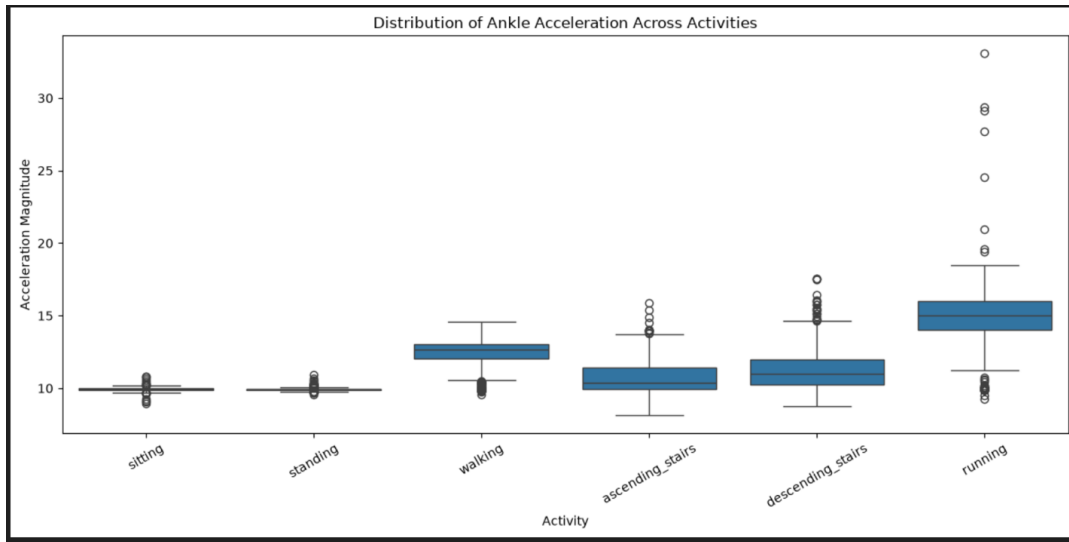


Figure 3. Distribution of ankle acceleration across activities. Sitting and standing form tight, low bands; running shows the widest spread and most extreme outliers.

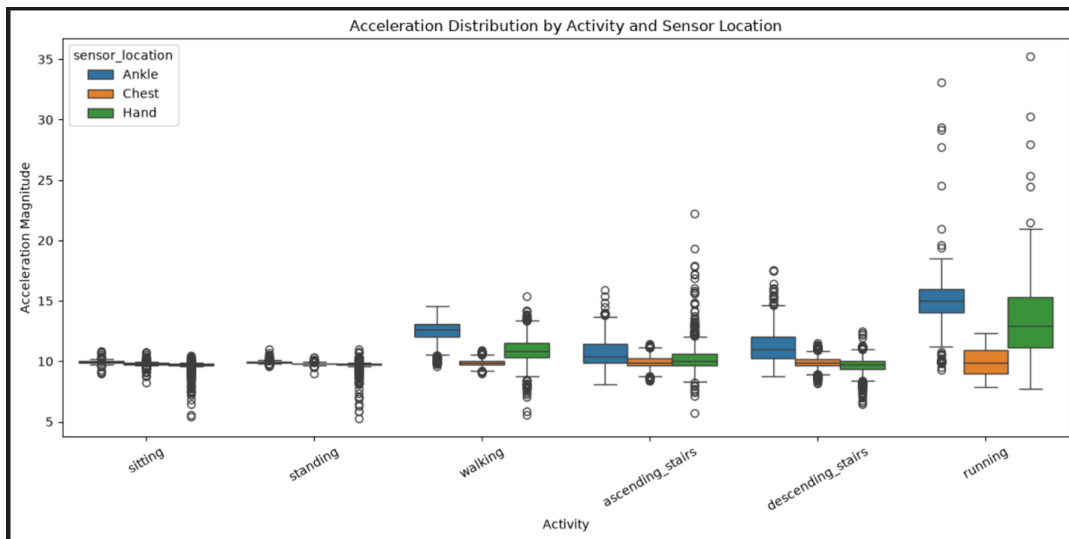


Figure 4. Acceleration distribution by activity and sensor location. Ankle (blue) consistently separates activities more clearly than chest (orange); hand (green) falls in between.

4.3 Consistency vs. Chaos: A Variability Analysis

Averages describe a "typical" moment, but standard deviation reveals how consistent or erratic an activity looks moment to moment — a second, independent lens on the same data.

Table 5. Standard Deviation of Acceleration Magnitude and Heart Rate by Activity

Activity	Ankle StdDev	Chest StdDev	Hand StdDev	Heart Rate StdDev
Sitting	0.12	0.12	0.37	8.02
Standing	0.08	0.09	0.30	10.01

Activity	Ankle StdDev	Chest StdDev	Hand StdDev	Heart Rate StdDev
Walking	0.87	0.29	0.93	9.54
Ascending Stairs	1.15	0.50	1.16	20.99
Descending Stairs	1.31	0.47	0.66	22.97
Running	2.14	1.07	2.92	22.88

Sitting and standing aren't just low in intensity — they're also remarkably consistent, with tiny standard deviations across every sensor. Running is both the most intense activity and the most erratic, which makes physical sense: different people run at very different paces and strides, while sitting still looks basically the same for everyone.

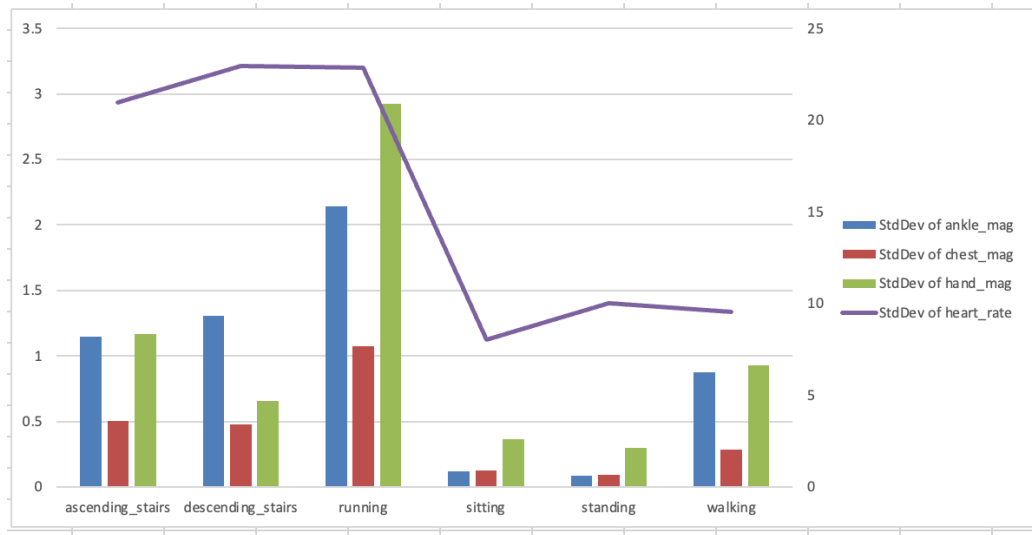


Figure 5. Variability (standard deviation) of sensor magnitude and heart rate by activity. Running's volatility stands apart; static postures barely fluctuate at all.

4.4 The Heart Rate Angle

Motion sensors measure what the body is doing; heart rate measures what that's costing the body physiologically — and the two don't always agree on which activities look "different."

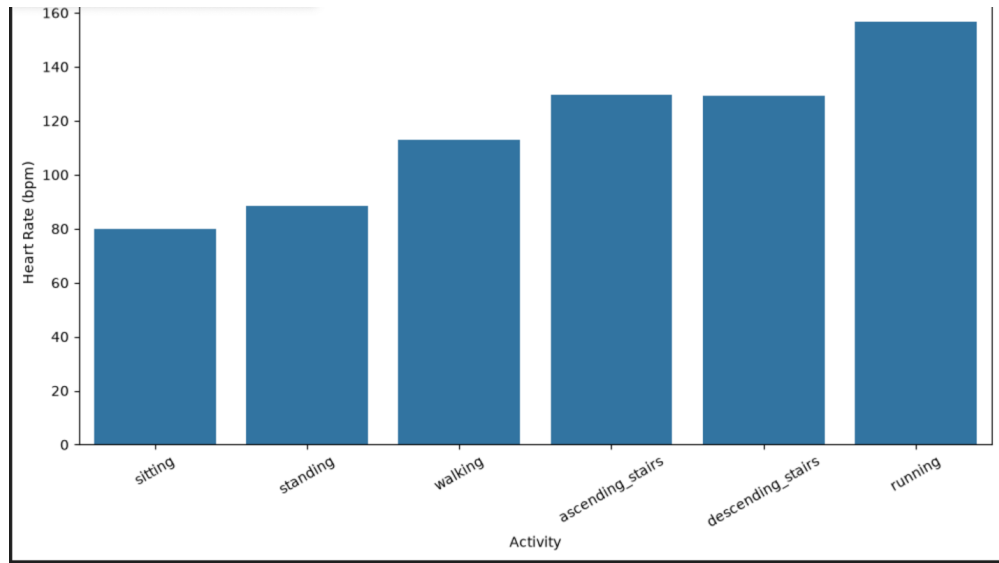


Figure 6. Average heart rate by activity. Heart rate cleanly separates sitting from standing — something no accelerometer in this study could do.

Two findings stand out here. First, sitting (80.0 bpm) and standing (88.5 bpm) are nearly identical on every accelerometer but clearly different in heart rate — meaning heart rate fills exactly the gap motion sensors leave behind. Second, ascending stairs (129.6 bpm) and descending stairs (129.2 bpm) show almost identical heart rate, despite the original data collection protocol rating ascending as far more physically demanding (8.0 METs) than descending (3.0 METs). The likely explanation: heart rate takes time to rise and fall, and both stair activities were performed back-to-back in only — roughly a minute each — so descending’s reading may simply reflect residual elevation from the preceding climb, rather than its own true demand. This is a meaningful limitation to flag rather than a contradiction to explain away.



Figure 7. Relationship between ankle acceleration and heart rate, colored by activity. Sitting and standing cluster tightly at low values on both axes; running spreads widest on both, showing the two signals are related but not redundant.

4.5 Can a Machine Learn to Tell These Activities Apart?

As a validation step, the raw signals were sliced into 2.56-second windows and summarized into 102 statistical features (mean, standard deviation, min, max, range) per window, per sensor. A Random Forest classifier was then trained and tested using a leave-one-subject-out approach — trained on 7 people, tested on the 8th, rotated across all 8 — to fairly estimate how well this would generalize to someone the model has never seen.

The result: 91% overall accuracy on subjects excluded from training.

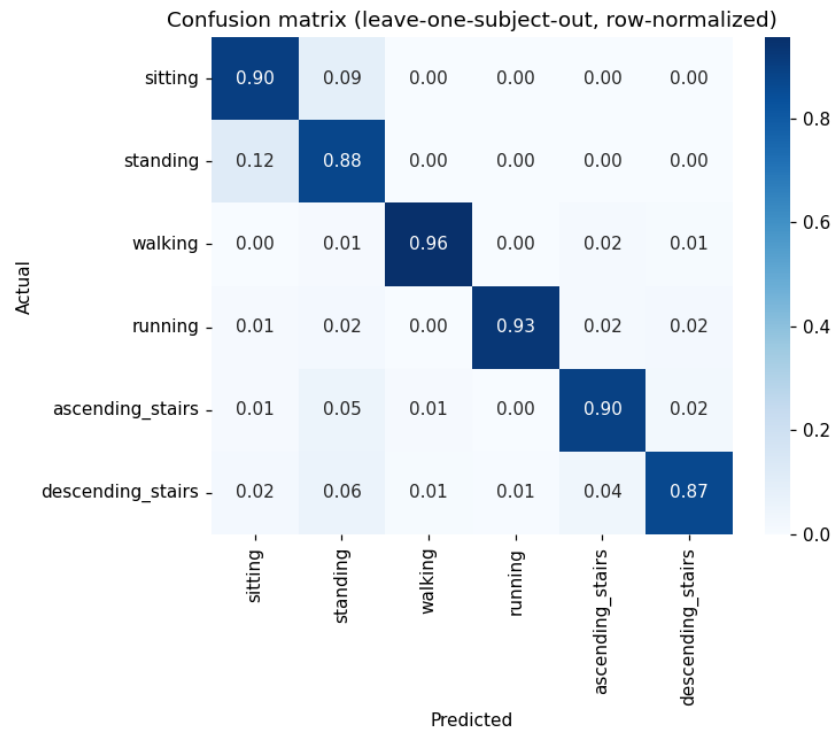


Figure 8. Confusion matrix (leave-one-subject-out, row-normalized). Walking and running are classified almost perfectly; sitting and standing are the most frequently confused pair, exactly as the earlier averages predicted.

Table 6. Classification Performance by Activity

Activity	Precision	Recall	F1-Score	Windows
Sitting	0.86	0.90	0.88	1,435
Standing	0.83	0.88	0.85	1,472
Walking	0.99	0.96	0.97	1,853
Running	0.99	0.93	0.96	759
Ascending Stairs	0.90	0.90	0.90	892
Descending Stairs	0.92	0.87	0.89	794

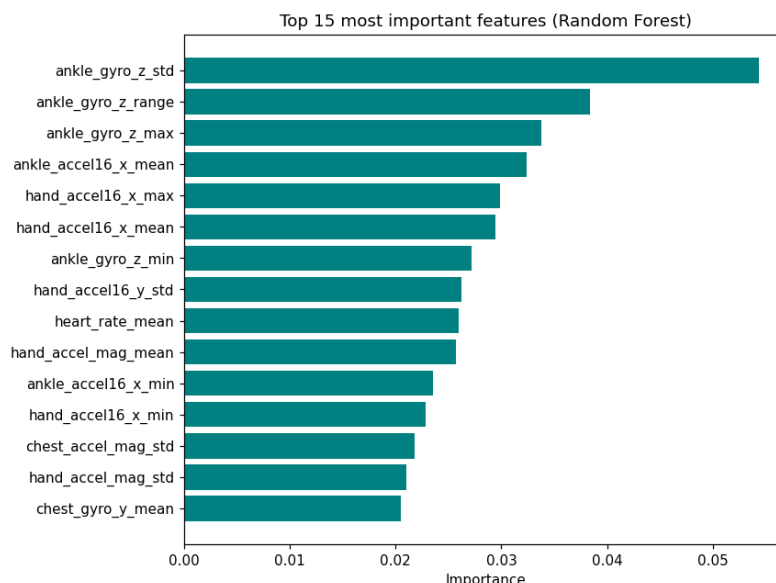


Figure 9. Top 15 most important features. Ankle rotation (gyroscope) ranks above every raw accelerometer feature — a non-obvious insight that rotation, not just movement magnitude, is what most reliably signals a gait cycle.

The model’s mistakes line up precisely with the earlier statistical findings: sitting and standing are confused with each other far more than with any other activity, confirming that motion-based classification alone genuinely cannot separate these two states — heart rate remains the more reliable signal there.

5. Share — Turning Numbers into a Story

Two visualization choices in this report were deliberate rather than default. First, wherever heart rate (roughly 80–160 bpm) is charted alongside accelerometer magnitude (roughly 9–15), the two are split across a primary and secondary axis — otherwise heart rate’s larger scale visually flattens the accelerometer differences into what looks like a flat line, hiding a real and meaningful pattern. Second, every activity comparison uses the same six-activity order throughout the report (sitting, standing, walking, running, ascending stairs, descending stairs, generally low-to-high intensity) so a reader’s eye doesn’t have to reorient between figures.

This document itself — combining Excel pivot-table analysis with Python-based statistical modeling and visualization — is the Share deliverable: a single narrative connecting a raw sensor dataset to an engineering recommendation.

6. Act — Engineering Recommendations

If designing a wearable device to detect these six activities, this analysis supports the following recommendations:

- Prioritize ankle placement for the primary motion sensor. It shows the widest range across activities (4.95 vs. 3.69 for hand and just 0.15 for chest) and drove the single most important feature in the classification model.
- Do not rely on a chest-only accelerometer for activity detection. Its near-flat response across all six activities (range of just 0.15) makes it the weakest of the three locations tested.
- Pair motion sensing with heart rate monitoring, not as a redundant backup but as a complementary signal — it is the only signal in this study that reliably separates sitting from standing.
- Treat ascending vs. descending stairs as a known weak point. Both movement and heart rate signals show meaningful overlap between these two activities; a production system may need a barometer (elevation change) or a longer observation window to reliably tell them apart.
- Validate beyond the lab. Every recommendation above comes from a controlled protocol with a narrow demographic (see Limitations) — real-world, free-living testing is a necessary next step before shipping any of these design decisions.

7. Limitations

- Small, skewed sample: 8 usable subjects, 8 of them male, all aged 23–32 — findings may not generalize to older adults, children, or a more balanced population.
- Lab-controlled protocol, not free-living data: subjects performed activities on cue in a structured sequence, which may not reflect how these activities look in ordinary daily life.
- Heart rate lag confound: back-to-back activities (like the two stair segments) may show heart rate still catching up from the previous activity rather than reflecting the current one.
- Dataset age: collected in 2012 with hardware of that era; modern wearable sensors may capture cleaner or higher-resolution signals.
- Subject 109 excluded entirely, reducing the effective sample for this specific six-activity comparison from 9 to 8.

8. Tools & Skills Demonstrated

Tools

- Excel / Google Sheets: pivot tables, dual-axis charting
- Python: pandas, scikit-learn, matplotlib, seaborn

Skills

- Data cleaning and validation
- Exploratory data analysis

- Statistical analysis (mean, standard deviation, range comparison)
- Feature engineering (windowing, signal magnitude, cross-sensor features)
- Data visualization (dual-axis charts, boxplots, confusion matrices)
- Data storytelling
- Engineering interpretation and recommendation-writing